



Experiment 1

Name : Arjun Dubey

UID : 23BCS13190

Branch: BE-CSE

Section/Group: KRG-3B

Semester: 6th

Date of Performance: 12-01-2026

Subject Name: Full Stack-II

Subject Code: 23CSH-309

1. Aim: To design and implement the foundational frontend architecture of the EcoTrack application using modern React practices, Vite tooling, and ES6+ JavaScript features.

2. Objective:-

- To understand about basic of React and Vite.
- To create a project using Vite with proper flow.
- To apply ES6 array methods (map, filter, reduce) for data-driven UI rendering
- To separate concerns using components, pages, and data modules

3. Implementation/Code:

- **Logs.js :-**
-

```
export const logs = [  
  { id: 1, activity: "Car Travel", carbon: 8 },  
  { id: 2, activity: "Electricity Usage", carbon: 6 },  
  { id: 3, activity: "Cycling", carbon: 0 },  
  {id: 4, activity: "Bus",carbon : 3},  
  {id: 5, activity: "Walking",carbon:0}  
];
```

- **Dashboard.jsx :-**

```
import { logs } from "../data/log";

const Dashboard = () => {
  const totalCarbon = logs.reduce((sum, log) => {
    return sum + log.carbon;
  }, 0);

  return (
    <div>
      <h2>Dashboard</h2>
      <p>Total Carbon Footprint: {totalCarbon} kgs</p>

      <ul>
        {logs.map((log) => (
          <li key={log.id}>
            {log.activity} = {log.carbon} Kg
          </li>
        ))}
      </ul>
    </div>
  );
};
export default Dashboard;
```

- **Logs.jsx :-**

```
import React from 'react'
import { logs } from '../data/log'

export const Logs = () => {
  const highCarbon = logs.filter(
    log => log.carbon >= 4
  )

  return (
    <div>
      <h2>High Carbon Activities more than 4</h2>

      {highCarbon.map((log, index) => (
        <p key={index}>
          {log.activity} - {log.carbon} kg
        </p>
      ))}
    </div>
  )
}

export const LowCarbon = () => {
  const LowCarbon = logs.filter(
    logs => logs.carbon <= 3
  )

  return (
    <div>
      <h2>Low Carbon Activities less than 3</h2>

      {logs.map((log, index) => {
        return (
          <p>
            {log.activity} - {log.carbon} kg
          </p>
        )
      })}
    </div>
  )
}
```



- App.jsx:-

```
import Header from "../components/header";
import Dashboard from "../pages/dashboard";
import Logs from "../pages/logs";
const App = () => {
  return (
    <>
      <Header title="EcoTrack Experiment 1" />
      <Dashboard />
      <Logs />
    </>
  );
};
export default App;
```

4. Output

The screenshot shows the output of the application. It features a green header with the title "EcoTrack Experiment 1". Below the header is a dark grey section titled "Dashboard". Under the dashboard title, it displays "Total Carbon Footprint: 17 kgs" followed by a bulleted list of activities and their carbon footprints: Car Travel = 8 Kg, Electricity Usage = 6 Kg, Cycling = 0 Kg, Bus = 3 Kg, and Walking = 0 Kg. Below the dashboard section is another dark grey section titled "Daily Logs". Under the logs title, it displays a bulleted list of activities and their carbon footprints: Car Travel = 8 kg, Electricity Usage = 6 kg, Cycling = 0 kg, Bus = 3 kg, and Walking = 0 kg. The text in the "Daily Logs" section is color-coded: "Car Travel" and "Bus" are red, "Electricity Usage" and "Walking" are green, and "Cycling" is black.

EcoTrack Experiment 1

Dashboard

Total Carbon Footprint: 17 kgs

- Car Travel = 8 Kg
- Electricity Usage = 6 Kg
- Cycling = 0 Kg
- Bus = 3 Kg
- Walking = 0 Kg

Daily Logs

- Car Travel = 8 kg
- Electricity Usage = 6 kg
- Cycling = 0 kg
- Bus = 3 kg
- Walking = 0 kg



5. Learning Outcome :-

- Created an Eco Tracker application using React to track carbon emissions.
- Displayed emission data through a dashboard built with reusable components.
- Applied JavaScript array functions to separate low and high emission activities.
- Organized the project using a component-based structure.
- Implemented dynamic rendering based on carbon emission values.
- Gained hands-on experience with data processing in React.